

Statistical Mechanics of Money:

Applying Boltzmann-Gibbs Distributions and Entropy
to Economic Dynamics via Agent-Based Simulation

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Abstract

2 **Keywords:** Statistical mechanics, Boltzmann-
2 Gibbs distribution, Gibbs entropy, Agent-based
simulation, Markov processes, Economic dynamics,
3 Complex systems.
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1 Introduction

The applicability of **statistical mechanics** to economic systems, specifically how money distribution in a society of interacting economic agents follows the **Boltzmann-Gibbs distribution**. [1, 2] This is a foundational principle from thermodynamics and statistical physics. The assumption is that money, like energy in a physical system, is conserved during transactions between agents.

I conducted agent based simulations modeling economic transactions as random exchanges of money between agents, analogous to particle collisions in a gas. Under various transaction mechanisms such as constant exchanges between agents, fraction γ of individual wealth per agent in each transaction, fraction of system average, etc., the simulated money distribution converges to an exponential form $P(m) = \frac{1}{T}e^{-m/T}$, where $T = M/N$ represents the system's effective "temperature" (average wealth per agent).

Two simulation models are analyzed:

1. **Basic money distribution** with no debt constraints
2. **System with debt** allowing limited negative balances

It's possible to demonstrate that **Gibbs entropy** correctly predicts equilibrium states and that the system evolves as a **Markov process** toward maximum entropy. These results validate the hypothesis that economic inequality and wealth concentration emerge as natural equilibrium states when agents interact via random transactions, analogously to how physical systems spontaneously approach thermodynamic equilibrium.

This research connects fundamental concepts from statistical mechanics from Markov processes, Boltzmann-Gibbs distributions, Gibbs entropy, and dynamical systems theory to economics and the so called "econophysics" or "social physics", providing a computational validation of theoretical predictions.

1.1 Motivation

The term "econophysics" was introduced by analogy with similar terms, such as astrophysics, geophysics, and biophysics, which describe applications of physics to different fields.

It should be emphasized that econophysics does not literally apply the laws of physics, such as Newton's laws or quantum mechanics, to humans, but rather uses mathematical methods developed in statistical physics to study properties of complex economic systems consisting of a large number of humans. So, it may be considered as a branch of applied theory of probabilities. Econophysics distances itself from the verbose, narrative, and ideological style of political economy and is closer to econometrics in its focus. Studying mathematical models of a large number of interacting economic agents, econophysics has much common ground with the agent-based modeling and simulation [3].

2 Model and Theory

2.1 Theoretical Framework

[Model and theory content to be filled...]

3 Results

3.1 Scenario 1: Basic Money Distribution

[Results content to be filled with figures and analysis...]

4 Conclusion

4.1 Summary

[Conclusion content to be filled...]

A Mathematical Details

A.1 Boltzmann-Gibbs Distribution

[Mathematical details to be filled...]

B Code Snippets

B.1 Core Simulation

[Code snippets to be filled...]

References

- [1] Anirban Chakraborti and Bikas K. Chakrabarti. Statistical mechanics of money: how saving propensity affects its distribution. *The European Physical Journal B*, 17(1):167–170, 2000.
- [2] Adrian Dragulescu and Victor M Yakovenko. Statistical mechanics of money. *The European Physical Journal B-Condensed Matter and Complex Systems*, 17(4):723–729, 2000.
- [3] Victor M. Yakovenko and J. Barkley Rosser. Colloquium: Statistical mechanics of money, wealth, and income. *Reviews of Modern Physics*, 81(4):1703–1725, 2009.